

ALFAAS33297

## Dichloroacetic acid, 99%

### SECTION 1. IDENTIFICATION OF THE SUBSTANCE/MIXTURE AND OF THE COMPANY/UNDERTAKING

|                                             |                                                                                                                                                                                                                                                                    |
|---------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>产品说明:</b><br><b>Product Description:</b> | <b>Dichloroacetic acid, 99%</b><br><b>Dichloroacetic acid, 99%</b>                                                                                                                                                                                                 |
| <b>Cat No. :</b>                            | <b>S33297</b>                                                                                                                                                                                                                                                      |
| <b>Synonyms</b>                             | Dichloroethanoic Acid; 2,2-Dichloroacetic Acid; Dichloroacetic Acid                                                                                                                                                                                                |
| <b>CAS No</b>                               | 79-43-6                                                                                                                                                                                                                                                            |
| <b>Molecular Formula</b>                    | C2 H2 Cl2 O2                                                                                                                                                                                                                                                       |
| <b>Supplier</b>                             | Avocado Research Chemicals Ltd.<br>(Part of Thermo Fisher Scientific)<br>Shore Road, Heysham<br>Lancashire, LA3 2XY,<br>United Kingdom<br>Office Tel: +44 (0) 1524 850506<br>Office Fax: +44 (0) 1524 850608                                                       |
| <b>Emergency Telephone Number</b>           | For information <b>US</b> call: 001-800-227-6701 / <b>Europe</b> call: +32 14 57 52 11<br>Emergency Number <b>US</b> :001-201-796-7100 / <b>Europe</b> : +32 14 57 52 99<br><b>CHEMTREC</b> Tel. No. <b>US</b> :001-800-424-9300 / <b>Europe</b> :001-703-527-3887 |
| <b>E-mail address</b>                       | begel.sdsdesk@thermofisher.com                                                                                                                                                                                                                                     |
| <b>Recommended Use</b>                      | Laboratory chemicals.                                                                                                                                                                                                                                              |
| <b>Uses advised against</b>                 | No Information available                                                                                                                                                                                                                                           |

### SECTION 2. HAZARD IDENTIFICATION

|                                                                                                                                                                                                                                                                                                                                                                                                                                          |                                   |                        |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------|------------------------|
| <b>Physical State</b><br>Liquid                                                                                                                                                                                                                                                                                                                                                                                                          | <b>Appearance</b><br>Light yellow | <b>Odor</b><br>pungent |
| <b>Emergency Overview</b><br>Causes severe skin burns and eye damage. Suspected of causing cancer. Very toxic to aquatic life. May be corrosive to metals. May be harmful if swallowed. Toxic in contact with skin. May damage fertility or the unborn child. Effects on or via lactation. May cause harm to breast-fed children. May cause damage to organs through prolonged or repeated exposure. Corrosive to the respiratory tract. |                                   |                        |

#### Classification of the substance or mixture

|                                                      |              |
|------------------------------------------------------|--------------|
| Substances/mixtures corrosive to metal               | Category 1   |
| Acute Oral Toxicity                                  | Category 5   |
| Acute Dermal Toxicity                                | Category 3   |
| Skin Corrosion/Irritation                            | Category 1 A |
| Serious Eye Damage/Eye Irritation                    | Category 1   |
| Carcinogenicity                                      | Category 2   |
| Reproductive Toxicity                                | Category 1B  |
| Effects on or via lactation                          |              |
| Specific target organ toxicity - (repeated exposure) | Category 2   |
| Acute aquatic toxicity                               | Category 1   |

#### Label Elements

## Dichloroacetic acid, 99%



## Signal Word

## Danger

## Hazard Statements

H290 - May be corrosive to metals  
H314 - Causes severe skin burns and eye damage  
H351 - Suspected of causing cancer  
H400 - Very toxic to aquatic life  
H303 - May be harmful if swallowed  
H311 - Toxic in contact with skin  
H360 - May damage fertility or the unborn child  
H362 - May cause harm to breast-fed children  
H373 - May cause damage to organs through prolonged or repeated exposure

## Precautionary Statements

## Prevention

P201 - Obtain special instructions before use  
P202 - Do not handle until all safety precautions have been read and understood  
P234 - Keep only in original packaging  
P260 - Do not breathe dust/fume/gas/mist/vapors/spray  
P263 - Avoid contact during pregnancy and while nursing  
P264 - Wash face, hands and any exposed skin thoroughly after handling  
P270 - Do not eat, drink or smoke when using this product  
P271 - Use only outdoors or in a well-ventilated area  
P284 - Wear respiratory protection  
P280 - Wear protective gloves/protective clothing/eye protection/face protection

## Response

P390 - Absorb spillage to prevent material damage  
P301 + P330 + P331 - IF SWALLOWED: rinse mouth. Do NOT induce vomiting  
P303 + P361 + P353 - IF ON SKIN (or hair): Take off immediately all contaminated clothing. Rinse skin with water or shower  
P304 + P340 - IF INHALED: Remove person to fresh air and keep comfortable for breathing  
P305 + P351 + P338 - IF IN EYES: Rinse cautiously with water for several minutes. Remove contact lenses, if present and easy to do. Continue rinsing  
P310 - Immediately call a POISON CENTER or doctor  
P362 + P364 - Take off contaminated clothing and wash it before reuse

## Storage

P402 - Store in a dry place  
P403 + P233 - Store in a well-ventilated place. Keep container tightly closed  
P406 - Store in corrosion resistant polypropylene container with a resistant liner  
P405 - Store locked up

## Disposal

P501 - Dispose of contents/ container to an approved waste disposal plant

## Physical and Chemical Hazards

May be corrosive to metals.

## Health Hazards

Corrosive. Causes skin and eye burns. Causes serious eye damage. Suspected of causing cancer. May be harmful if swallowed. Toxic in contact with skin. May damage fertility or the unborn child. May cause harm to breast-fed children. May cause damage to organs through prolonged or repeated exposure.

## Environmental hazards

Very toxic to aquatic life. . Will likely be mobile in the environment due to its water solubility. The product is water soluble, and may spread in water systems.

This product does not contain any known or suspected endocrine disruptors.

## SECTION 3. COMPOSITION/INFORMATION ON INGREDIENTS

**SAFETY DATA SHEET**

Dichloroacetic acid, 99%

| Component           | CAS No  | Weight % |
|---------------------|---------|----------|
| Dichloroacetic acid | 79-43-6 | >95      |

**SECTION 4. FIRST AID MEASURES****General Advice**

Show this safety data sheet to the doctor in attendance. Immediate medical attention is required.

**Eye Contact**

Rinse immediately with plenty of water, also under the eyelids, for at least 15 minutes. In the case of contact with eyes, rinse immediately with plenty of water and seek medical advice.

**Skin Contact**

Wash off immediately with plenty of water for at least 15 minutes. Immediate medical attention is required.

**Inhalation**

If not breathing, give artificial respiration. Do not use mouth-to-mouth method if victim ingested or inhaled the substance; give artificial respiration with the aid of a pocket mask equipped with a one-way valve or other proper respiratory medical device. Remove to fresh air. Immediate medical attention is required.

**Ingestion**

Do NOT induce vomiting. Call a physician or poison control center immediately.

**Most important symptoms and effects**

Causes burns by all exposure routes. Product is a corrosive material. Use of gastric lavage or emesis is contraindicated. Possible perforation of stomach or esophagus should be investigated: Ingestion causes severe swelling, severe damage to the delicate tissue and danger of perforation

**Self-Protection of the First Aider**

Ensure that medical personnel are aware of the material(s) involved, take precautions to protect themselves and prevent spread of contamination.

**Notes to Physician**

Treat symptomatically.

**SECTION 5. FIRE-FIGHTING MEASURES****Suitable Extinguishing Media**

CO<sub>2</sub>, dry chemical, dry sand, alcohol-resistant foam.

**Extinguishing media which must not be used for safety reasons**

No information available.

**Specific Hazards Arising from the Chemical**

Thermal decomposition can lead to release of irritating gases and vapors. The product causes burns of eyes, skin and mucous membranes. Do not allow run-off from fire-fighting to enter drains or water courses.

**Protective Equipment and Precautions for Firefighters**

As in any fire, wear self-contained breathing apparatus pressure-demand, MSHA/NIOSH (approved or equivalent) and full protective gear. Thermal decomposition can lead to release of irritating gases and vapors.

**SECTION 6. ACCIDENTAL RELEASE MEASURES****Personal Precautions**

Ensure adequate ventilation. Use personal protective equipment as required. Evacuate personnel to safe areas. Keep people away from and upwind of spill/leak.

## Dichloroacetic acid, 99%

**Environmental Precautions**

Do not flush into surface water or sanitary sewer system. Do not allow material to contaminate ground water system. Prevent product from entering drains. Local authorities should be advised if significant spillages cannot be contained.

**Methods for Containment and Clean Up**

Soak up with inert absorbent material. Keep in suitable, closed containers for disposal.

Refer to protective measures listed in Sections 8 and 13.

**SECTION 7. HANDLING AND STORAGE****Handling**

Wear personal protective equipment/face protection. Do not get in eyes, on skin, or on clothing. Use only under a chemical fume hood. Do not breathe mist/vapors/spray. Do not ingest. If swallowed then seek immediate medical assistance.

**Storage**

Keep in a dry, cool and well-ventilated place. Keep container tightly closed. To maintain product quality: Store under an inert atmosphere.

**Specific Use(s)**

Use in laboratories

**SECTION 8. EXPOSURE CONTROLS/PERSONAL PROTECTION****Control Parameters**

| Component           | ACGIH TLV            | OSHA PEL | NIOSH | The United Kingdom | European Union |
|---------------------|----------------------|----------|-------|--------------------|----------------|
| Dichloroacetic acid | TWA: 0.5 ppm<br>Skin |          |       | -                  |                |

Legend

ACGIH - American Conference of Governmental Industrial Hygienists

**Exposure Controls****Engineering Measures**

Use only under a chemical fume hood. Ensure adequate ventilation, especially in confined areas. Ensure that eyewash stations and safety showers are close to the workstation location. Wherever possible, engineering control measures such as the isolation or enclosure of the process, the introduction of process or equipment changes to minimise release or contact, and the use of properly designed ventilation systems, should be adopted to control hazardous materials at source.

**Personal protective equipment****Eye Protection**

Goggles (European standard - EN 166)

**Hand Protection**

Protective gloves

| Glove material | Breakthrough time | Glove thickness | EU standard | Glove comments        |
|----------------|-------------------|-----------------|-------------|-----------------------|
| Natural rubber | See manufacturers | -               | EN 374      | (minimum requirement) |
| Butyl rubber   | recommendations   |                 |             |                       |
| Nitrile rubber |                   |                 |             |                       |
| Neoprene       |                   |                 |             |                       |
| PVC            |                   |                 |             |                       |

Inspect gloves before use.

Please observe the instructions regarding permeability and breakthrough time which are provided by the supplier of the gloves. (Refer to manufacturer/supplier for information)

Ensure gloves are suitable for the task: Chemical compatibility, Dexterity, Operational conditions, User susceptibility, e.g. sensitisation effects, also take into consideration the specific local conditions under which the product is used, such as the danger

## Dichloroacetic acid, 99%

of cuts, abrasion.

Remove gloves with care avoiding skin contamination.

|                                        |                                                                                                                                                                                                                                                                                                                             |
|----------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Skin and body protection</b>        | Long sleeved clothing                                                                                                                                                                                                                                                                                                       |
| <b>Respiratory Protection</b>          | When workers are facing concentrations above the exposure limit they must use appropriate certified respirators.<br>To protect the wearer, respiratory protective equipment must be the correct fit and be used and maintained properly                                                                                     |
| <b>Large scale/emergency use</b>       | Use a NIOSH/MSHA or European Standard EN 136 approved respirator if exposure limits are exceeded or if irritation or other symptoms are experienced<br><b>Recommended Filter type:</b> Particulates filter conforming to EN 143 Acid gases filter Type E Yellow conforming to EN14387                                       |
| <b>Small scale/Laboratory use</b>      | Use a NIOSH/MSHA or European Standard EN 149:2001 approved respirator if exposure limits are exceeded or if irritation or other symptoms are experienced.<br><b>Recommended half mask:-</b> Valve filtering: EN405; or; Half mask: EN140; plus filter, EN 141<br>When RPE is used a face piece Fit Test should be conducted |
| <b>Hygiene Measures</b>                | Handle in accordance with good industrial hygiene and safety practice.                                                                                                                                                                                                                                                      |
| <b>Environmental exposure controls</b> | Prevent product from entering drains. Do not allow material to contaminate ground water system. Local authorities should be advised if significant spillages cannot be contained.                                                                                                                                           |

## SECTION 9. PHYSICAL AND CHEMICAL PROPERTIES

|                                                |                            |                                          |
|------------------------------------------------|----------------------------|------------------------------------------|
| <b>Appearance</b>                              | Light yellow               |                                          |
| <b>Physical State</b>                          | Liquid                     |                                          |
| <b>Odor</b>                                    | pungent                    |                                          |
| <b>Odor Threshold</b>                          | No data available          |                                          |
| <b>pH</b>                                      | 1.2                        | 129 g/l                                  |
| <b>Melting Point/Range</b>                     | 9 - 11 °C / 48.2 - 51.8 °F |                                          |
| <b>Softening Point</b>                         | No data available          |                                          |
| <b>Boiling Point/Range</b>                     | 194 °C / 381.2 °F          | @ 760 mmHg                               |
| <b>Flash Point</b>                             | > 112 °C / > 233.6 °F      | <b>Method -</b> No information available |
| <b>Evaporation Rate</b>                        | No data available          |                                          |
| <b>Flammability (solid,gas)</b>                | Not applicable             | Liquid                                   |
| <b>Explosion Limits</b>                        | No data available          |                                          |
| <b>Vapor Pressure</b>                          | 1.3 mbar @ 44 °C           |                                          |
| <b>Vapor Density</b>                           | 4.45                       | (Air = 1.0)                              |
| <b>Specific Gravity / Density</b>              | 1.560                      |                                          |
| <b>Bulk Density</b>                            | Not applicable             | Liquid                                   |
| <b>Water Solubility</b>                        | Soluble                    |                                          |
| <b>Solubility in other solvents</b>            | No information available   |                                          |
| <b>Partition Coefficient (n-octanol/water)</b> |                            |                                          |
| <b>Component</b>                               | <b>log Pow</b>             |                                          |
| Dichloroacetic acid                            | 0.942                      |                                          |
| <b>Autoignition Temperature</b>                | 194 °C / 381.2 °F          |                                          |
| <b>Decomposition Temperature</b>               | No data available          |                                          |
| <b>Viscosity</b>                               | No data available          |                                          |
| <b>Explosive Properties</b>                    | No information available   |                                          |
| <b>Oxidizing Properties</b>                    | No information available   |                                          |
| <b>Molecular Formula</b>                       | C2 H2 Cl2 O2               |                                          |
| <b>Molecular Weight</b>                        | 128.94                     |                                          |

## SECTION 10. STABILITY AND REACTIVITY

|                                         |                                                                                           |
|-----------------------------------------|-------------------------------------------------------------------------------------------|
| <b>Stability</b>                        | Stable under recommended storage conditions.                                              |
| <b>Hazardous Reactions</b>              | None under normal processing.                                                             |
| <b>Hazardous Polymerization</b>         | Hazardous polymerization does not occur.                                                  |
| <b>Conditions to Avoid</b>              | Incompatible products. Excess heat.                                                       |
| <b>Materials to avoid</b>               | Strong oxidizing agents. Strong bases. Strong reducing agents. Metals. .                  |
| <b>Hazardous Decomposition Products</b> | Carbon monoxide (CO). Carbon dioxide (CO <sub>2</sub> ). Phosgene. Hydrogen chloride gas. |

## SECTION 11. TOXICOLOGICAL INFORMATION

## Product Information

## (a) acute toxicity;

| Component           | LD50 Oral                 | LD50 Dermal                 | LC50 Inhalation |
|---------------------|---------------------------|-----------------------------|-----------------|
| Dichloroacetic acid | LD50 = 2820 mg/kg ( Rat ) | LD50 = 510 mg/kg ( Rabbit ) |                 |

(b) skin corrosion/irritation; Category 1 A

(c) serious eye damage/irritation; Category 1

## (d) respiratory or skin sensitization;

Respiratory  
Skin

Based on available data, the classification criteria are not met

Based on available data, the classification criteria are not met

(e) germ cell mutagenicity; Based on available data, the classification criteria are not met

## (f) carcinogenicity;

Category 2

The table below indicates whether each agency has listed any ingredient as a carcinogen

| Component           | EU | UK | Germany | IARC     |
|---------------------|----|----|---------|----------|
| Dichloroacetic acid |    |    |         | Group 2B |

(g) reproductive toxicity; Category 1B

(h) STOT-single exposure; Based on available data, the classification criteria are not met

(i) STOT-repeated exposure; Category 2

Target Organs

Liver, Brain.

(j) aspiration hazard; Based on available data, the classification criteria are not met

## Other Adverse Effects

**Symptoms / effects, both acute and delayed** Product is a corrosive material. Use of gastric lavage or emesis is contraindicated. Possible perforation of stomach or esophagus should be investigated: Ingestion causes severe swelling, severe damage to the delicate tissue and danger of perforation

**SAFETY DATA SHEET**

Dichloroacetic acid, 99%

**SECTION 12. ECOLOGICAL INFORMATION****Ecotoxicity effects**

Very toxic to aquatic organisms. The product contains following substances which are hazardous for the environment.

| Component           | Freshwater Fish | Water Flea        | Freshwater Algae | Microtox |
|---------------------|-----------------|-------------------|------------------|----------|
| Dichloroacetic acid |                 | 106-2600 mg/L 24h |                  |          |

**Persistence and Degradability****Persistence**

Readily biodegradable

**Degradation in sewage treatment plant**

Persistence is unlikely.

Contains substances known to be hazardous to the environment or not degradable in waste water treatment plants.

**Bioaccumulative Potential**

Bioaccumulation is unlikely

| Component           | log Pow | Bioconcentration factor (BCF) |
|---------------------|---------|-------------------------------|
| Dichloroacetic acid | 0.942   | No data available             |

**Mobility in soil**

The product is water soluble, and may spread in water systems. Will likely be mobile in the environment due to its water solubility. Highly mobile in soils.

**Endocrine Disruptor Information**

This product does not contain any known or suspected endocrine disruptors.

**Persistent Organic Pollutant**

This product does not contain any known or suspected substance.

**Ozone Depletion Potential**

This product does not contain any known or suspected substance.

**SECTION 13. DISPOSAL CONSIDERATIONS****Waste from Residues/Unused Products**

Should not be released into the environment. Waste is classified as hazardous. Dispose of in accordance with the European Directives on waste and hazardous waste. Dispose of in accordance with local regulations.

**Contaminated Packaging**

Dispose of this container to hazardous or special waste collection point.

**Other Information**

Do not flush to sewer. Waste codes should be assigned by the user based on the application for which the product was used. Do not empty into drains. Large amounts will affect pH and harm aquatic organisms. Solutions with low pH-value must be neutralized before discharge. Do not let this chemical enter the environment.

**SECTION 14. TRANSPORT INFORMATION****Road and Rail Transport**

|                      |                     |
|----------------------|---------------------|
| UN-No                | UN1764              |
| Proper Shipping Name | DICHLOROACETIC ACID |
| Hazard Class         | 8                   |
| Packing Group        | II                  |

**IMDG/IMO**

|                      |                     |
|----------------------|---------------------|
| UN-No                | UN1764              |
| Proper Shipping Name | DICHLOROACETIC ACID |
| Hazard Class         | 8                   |
| Packing Group        | II                  |

**IATA**

**SAFETY DATA SHEET**

Dichloroacetic acid, 99%

UN-No UN1764  
 Proper Shipping Name DICHLOROACETIC ACID  
 Hazard Class 8  
 Packing Group II

Special Precautions for User No special precautions required

**SECTION 15. REGULATORY INFORMATION****International Inventories**

X = listed, China (IECSC), Europe (EINECS/ELINCS/NLP), U.S.A. (TSCA), Canada (DSL/NDSL), Philippines (PICCS), Japan (ENCS), Japan (ISHL), Australia (AICS), Korea (KECL).

| Component           | The Inventory of Hazardous Chemicals (2015 Edition) | List of dangerous goods GB 12268 - 2012 | TCSI | IECSC | EINECS    | TSCA | DSL | PICCS | ENCS | ISHL | AICS | KECL     |
|---------------------|-----------------------------------------------------|-----------------------------------------|------|-------|-----------|------|-----|-------|------|------|------|----------|
| Dichloroacetic acid | X                                                   | X                                       | X    | X     | 201-207-0 | X    | X   | X     | X    | X    | X    | KE-10054 |

**National Regulations****SECTION 16. OTHER INFORMATION**

Prepared By Health, Safety and Environmental Department  
 Creation Date 03-May-2012  
 Revision Date 13-May-2024  
 Revision Summary New emergency telephone response service provider.

**Training Advice**

Chemical incident response training.

**Legend**

**CAS** - Chemical Abstracts Service

**EINECS/ELINCS** - European Inventory of Existing Commercial Chemical Substances/EU List of Notified Chemical Substances

**PICCS** - Philippines Inventory of Chemicals and Chemical Substances

**IECSC** - Chinese Inventory of Existing Chemical Substances

**KECL** - Korean Existing and Evaluated Chemical Substances

**TSCA** - United States Toxic Substances Control Act Section 8(b) Inventory

**DSL/NDSL** - Canadian Domestic Substances List/Non-Domestic Substances List

**ENCS** - Japanese Existing and New Chemical Substances

**AICS** - Australian Inventory of Chemical Substances

**NZIoC** - New Zealand Inventory of Chemicals

**WEL** - Workplace Exposure Limit

**ACGIH** - American Conference of Governmental Industrial Hygienists

**DNEL** - Derived No Effect Level

**RPE** - Respiratory Protective Equipment

**LC50** - Lethal Concentration 50%

**NOEC** - No Observed Effect Concentration

**PBT** - Persistent, Bioaccumulative, Toxic

**TWA** - Time Weighted Average

**IARC** - International Agency for Research on Cancer

**PNEC** - Predicted No Effect Concentration

**LD50** - Lethal Dose 50%

**EC50** - Effective Concentration 50%

**POW** - Partition coefficient Octanol:Water

**vPvB** - very Persistent, very Bioaccumulative

**ICAO/IATA** - International Civil Aviation Organization/International Air Transport Association

**ADR** - European Agreement Concerning the International Carriage of Dangerous Goods by Road

**OECD** - Organisation for Economic Co-operation and Development

**BCF** - Bioconcentration factor

**IMO/IMDG** - International Maritime Organization/International Maritime Dangerous Goods Code

**MARPOL** - International Convention for the Prevention of Pollution from Ships

**ATE** - Acute Toxicity Estimate

**VOC** - (Volatile Organic Compound)



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**Key literature references and sources for data**

<https://echa.europa.eu/information-on-chemicals>

Suppliers safety data sheet, Chemadvisor - LOLI, Merck index, RTECS

**Disclaimer**

The information provided in this Safety Data Sheet is correct to the best of our knowledge, information and belief at the date of its publication. The information given is designed only as a guidance for safe handling, use, processing, storage, transportation, disposal and release and is not to be considered a warranty or quality specification. The information relates only to the specific material designated and may not be valid for such material used in combination with any other materials or in any process, unless specified in the text

**End of Safety Data Sheet**